

Discretising the aggregated covariance and differentiating through the scale parameter

The SDALGCP2 continuous- ϕ (“direct”) Monte Carlo likelihood

SDALGCP2 development notes

June 16, 2026

Abstract

The spatially discrete approximation to a log-Gaussian Cox process (SDA-LGCP) replaces the intractable areal integral of an LGCP by a finite sum over candidate points inside each region. The spatial *scale* parameter ϕ enters the model only through a *double integral* of the correlation function over pairs of regions. The original SDALGCP discretises ϕ on a grid and profiles over it. Here we (i) make explicit the double integral and the quadrature used to approximate it, (ii) differentiate *through* the integral so that ϕ can be estimated *continuously* inside the Monte Carlo maximum likelihood (MCML) objective, and (iii) record the gradient and Hessian that SDALGCP2 uses for the “direct” method. All derivatives are verified numerically.

1 Model and the aggregated random effect

Let $S(\cdot)$ be a zero-mean stationary Gaussian process on $\mathbb{R}^2$ with

$$\text{Cov}\{S(x), S(x')\} = \sigma^2 \rho(\|x - x'\|; \phi), \quad \rho(u; \phi) = \exp(-u/\phi), \quad (1)$$

the exponential (Matérn $\kappa = \frac{1}{2}$) correlation with scale $\phi > 0$. For an LGCP with intensity $\lambda(x) = \exp\{d(x)^\top \boldsymbol{\beta} + S(x)\}$, the SDA-LGCP models the count Y_i in region A_i ($i = 1, \dots, N$) as conditionally Poisson with a region-level linear predictor that is a *spatial average* of S ,

$$T_i(S) = \int_{A_i} S(x) w_i(x) dx, \quad w_i(x) \geq 0, \quad \int_{A_i} w_i(x) dx = 1. \quad (2)$$

Two weightings are used: the *uniform* average $w_i(x) = 1/|A_i|$, and the *population-weighted* average $w_i(x) = p(x)/\int_{A_i} p(u) du$, where $p(\cdot)$ is population density. Because (2) is a linear functional of the Gaussian field, the vector $\mathbf{S}^* = (T_1, \dots, T_N)^\top$ is Gaussian,

$$\mathbf{S}^* \sim \mathcal{N}(\mathbf{D}\boldsymbol{\beta}, \sigma^2 \mathbf{R}(\phi)), \quad Y_i \mid \mathbf{S}^* \sim \text{Poisson}(m_i e^{S_i^*}), \quad (3)$$

with $\mathbf{D}$ the design matrix and m_i an offset. Everything downstream operates on the $N \times N$ matrix $\mathbf{R}(\phi)$.

2 The scale parameter lives inside a double integral

By bilinearity of covariance and (1)–(2),

$$\boxed{\mathbf{R}_{ij}(\phi) = \text{Cov}\{T_i(S), T_j(S)\} / \sigma^2 = \int_{A_i} \int_{A_j} w_i(x) w_j(y) \exp\left(-\frac{\|x - y\|}{\phi}\right) dx dy} \quad (4)$$

This is the object of interest: ϕ appears *inside* a four-dimensional integral over the pair of regions $A_i \times A_j$, which has no closed form for general polygons. The diagonal term $\mathbf{R}_{ii}(\phi)$ is the same integral over $A_i \times A_i$ and is < 1 (it averages correlations ≤ 1 , equalling 1 only in the degenerate point-region limit).

3 Quadrature: discretising the double integral

Place candidate points $x_{i1}, \dots, x_{i,n_i} \in A_i$ with quadrature weights $\omega_{ik} \geq 0$, $\sum_k \omega_{ik} = 1$, chosen so that

$$\int_{A_i} f(x) w_i(x) dx \approx \sum_{k=1}^{n_i} \omega_{ik} f(x_{ik}). \quad (5)$$

For the uniform average with points drawn uniformly (or on a regular lattice), $\omega_{ik} = 1/n_i$; this is plain Monte Carlo / product-midpoint cubature. For the population-weighted average, the natural plug-in is the *ratio estimator*

$$\omega_{ik} = \frac{p(x_{ik})}{\sum_{l=1}^{n_i} p(x_{il})}, \quad (6)$$

evaluated from the population raster. Substituting (5) into (4) for both regions gives the computable approximation

$$\mathbf{R}_{ij}(\phi) \approx \hat{\mathbf{R}}_{ij}(\phi) = \sum_{k=1}^{n_i} \sum_{l=1}^{n_j} \omega_{ik} \omega_{jl} \exp\left(-\frac{d_{ik,jl}}{\phi}\right), \quad d_{ik,jl} = \|x_{ik} - x_{jl}\|. \quad (7)$$

Equation (7) is exactly what the C++ kernel `corr_aggregate_cpp` evaluates (each distance computed once, all of ϕ reduced in an inner loop, parallel over region pairs).

Accuracy. For uniform weights and n i.i.d. uniform points per region, $\hat{\mathbf{R}}_{ij}$ is an unbiased Monte Carlo estimator of $\mathbf{R}_{ij}$ with variance $O(1/n)$; for a regular lattice of spacing δ it is a product cubature with bias $O(\delta^2)$ away from the diagonal. The integrand $\exp(-\|x - y\|/\phi)$ is Lipschitz in its argument with constant $1/\phi$, so the approximation is accurate provided the point spacing satisfies $\delta \ll \phi$; the only non-smoothness is the cusp of $\|x - y\|$ at $x = y$, which affects the diagonal blocks ($i = j$) alone. This is the formal reason the candidate-point spacing δ must be chosen small relative to the spatial range.

4 Differentiating through the integral

Because $\phi \mapsto \exp(-u/\phi)$ is smooth on $(0, \infty)$ and, for u in a bounded region, is dominated together with its ϕ -derivatives by an integrable function, the Leibniz rule (dominated convergence) permits differentiation under the integral sign in (4). With

$$\frac{\partial}{\partial \phi} e^{-u/\phi} = e^{-u/\phi} \frac{u}{\phi^2}, \quad \frac{\partial^2}{\partial \phi^2} e^{-u/\phi} = e^{-u/\phi} \left(\frac{u^2}{\phi^4} - \frac{2u}{\phi^3} \right),$$

we obtain, writing $u = \|x - y\|$,

$$\dot{\mathbf{R}}_{ij}(\phi) \equiv \frac{\partial \mathbf{R}_{ij}}{\partial \phi} = \int_{A_i} \int_{A_j} w_i w_j e^{-u/\phi} \frac{u}{\phi^2} dx dy, \quad (8)$$

$$\ddot{\mathbf{R}}_{ij}(\phi) \equiv \frac{\partial^2 \mathbf{R}_{ij}}{\partial \phi^2} = \int_{A_i} \int_{A_j} w_i w_j e^{-u/\phi} \left(\frac{u^2}{\phi^4} - \frac{2u}{\phi^3} \right) dx dy, \quad (9)$$

each discretised by the same double sum (7) over candidate points. These are returned (alongside $\hat{\mathbf{R}}$) by `corr_and_grad_cpp`. Numerically, $\hat{\mathbf{R}}$ and $\hat{\hat{\mathbf{R}}}$ agree with central finite differences of $\hat{\mathbf{R}}$ to 2×10^{-11} and 8×10^{-6} respectively.

5 MCML with continuous ϕ

Let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \sigma^2, \phi)$. From (3), the joint log-density of $(\mathbf{S}^*, \mathbf{Y})$ splits into a data term (free of $\boldsymbol{\theta}$) and the Gaussian prior term

$$g(\mathbf{S}; \boldsymbol{\theta}) = -\frac{1}{2} \left[n \log \sigma^2 + \log |\mathbf{R}(\phi)| + \sigma^{-2} (\mathbf{S} - \mathbf{D}\boldsymbol{\beta})^\top \mathbf{R}(\phi)^{-1} (\mathbf{S} - \mathbf{D}\boldsymbol{\beta}) \right]. \quad (10)$$

Following Christensen (2004), draw $\mathbf{S}^{(1)}, \dots, \mathbf{S}^{(B)}$ from the conditional $[\mathbf{S}^* | \mathbf{Y}; \boldsymbol{\theta}_0]$ at an anchor $\boldsymbol{\theta}_0$ (here via the Laplace-preconditioned MALA sampler). Since the data term cancels in the ratio, the Monte Carlo log-likelihood (up to an additive constant) is

$$\ell_{\text{MC}}(\boldsymbol{\theta}) = \log \left\{ \frac{1}{B} \sum_{b=1}^B \exp[g(\mathbf{S}^{(b)}; \boldsymbol{\theta}) - g(\mathbf{S}^{(b)}; \boldsymbol{\theta}_0)] \right\}. \quad (11)$$

The *grid* method fixes ϕ at grid values and maximises (11) over $(\boldsymbol{\beta}, \sigma^2)$ at each; the *direct* method maximises (11) over all of $(\boldsymbol{\beta}, \log \sigma^2, \log \phi)$ jointly. We use the log parameterisation $\psi_s = \log \sigma^2$, $\psi_p = \log \phi$ to keep the variances positive.

5.1 Gradient

Write $\mathbf{u}_b = \mathbf{S}^{(b)} - \mathbf{D}\boldsymbol{\beta}$, $q_b = \mathbf{u}_b^\top \mathbf{R}^{-1} \mathbf{u}_b$, $\mathbf{R}^{-1} = \mathbf{R}(\phi)^{-1}$, $\dot{\mathbf{R}} = \dot{\mathbf{R}}(\phi)$, and the self-normalised importance weights $W_b = \exp\{a_b\} / \sum_c \exp\{a_c\}$ with $a_b = g(\mathbf{S}^{(b)}; \boldsymbol{\theta}) - g(\mathbf{S}^{(b)}; \boldsymbol{\theta}_0)$. Then $\nabla \ell_{\text{MC}} = \sum_b W_b \nabla g(\mathbf{S}^{(b)}; \boldsymbol{\theta})$, and from (10),

$$\frac{\partial g_b}{\partial \boldsymbol{\beta}} = \sigma^{-2} \mathbf{D}^\top \mathbf{R}^{-1} \mathbf{u}_b, \quad (12)$$

$$\frac{\partial g_b}{\partial \psi_s} = -\frac{n}{2} + \frac{q_b}{2\sigma^2}, \quad (13)$$

$$\frac{\partial g_b}{\partial \psi_p} = \phi \left[-\frac{1}{2} \text{tr}(\mathbf{R}^{-1} \dot{\mathbf{R}}) + \frac{1}{2\sigma^2} \mathbf{u}_b^\top \mathbf{R}^{-1} \dot{\mathbf{R}} \mathbf{R}^{-1} \mathbf{u}_b \right]. \quad (14)$$

The first two reproduce the fixed- ϕ gradient; (14) is the new term, obtained from $\partial_\phi \log |\mathbf{R}| = \text{tr}(\mathbf{R}^{-1} \dot{\mathbf{R}})$ and $\partial_\phi (\mathbf{u}^\top \mathbf{R}^{-1} \mathbf{u}) = -\mathbf{u}^\top \mathbf{R}^{-1} \dot{\mathbf{R}} \mathbf{R}^{-1} \mathbf{u}$, times ϕ for the log-chain rule.

5.2 Hessian

With $a_b = g_b(\boldsymbol{\theta}) - g_b(\boldsymbol{\theta}_0)$,

$$\nabla^2 \ell_{\text{MC}} = \sum_b W_b (\nabla^2 g_b + \nabla g_b \nabla g_b^\top) - \left(\sum_b W_b \nabla g_b \right) \left(\sum_b W_b \nabla g_b \right)^\top. \quad (15)$$

The blocks of $\nabla^2 g_b$, writing $\mathbf{A} = \mathbf{R}^{-1} \dot{\mathbf{R}}$, $\mathbf{M} = \mathbf{R}^{-1} \dot{\mathbf{R}} \mathbf{R}^{-1}$ and $\mathbf{n} = \mathbf{R}^{-1} (2\dot{\mathbf{R}} \mathbf{R}^{-1} \dot{\mathbf{R}} - \ddot{\mathbf{R}}) \mathbf{R}^{-1}$, are

$$\begin{aligned} \partial_{\boldsymbol{\beta}\boldsymbol{\beta}} &= -\sigma^{-2} \mathbf{D}^\top \mathbf{R}^{-1} \mathbf{D}, & \partial_{\boldsymbol{\beta}\psi_s} &= -\sigma^{-2} \mathbf{D}^\top \mathbf{R}^{-1} \mathbf{u}_b, \\ \partial_{\boldsymbol{\beta}\psi_p} &= -\phi \sigma^{-2} \mathbf{D}^\top \mathbf{M} \mathbf{u}_b, & \partial_{\psi_s \psi_s} &= -\frac{q_b}{2\sigma^2}, \\ \partial_{\psi_s \psi_p} &= -\frac{\phi}{2\sigma^2} \mathbf{u}_b^\top \mathbf{M} \mathbf{u}_b, & \partial_{\psi_p \psi_p} &= \phi^2 \left[t_2 - \frac{1}{2\sigma^2} \mathbf{u}_b^\top \mathbf{n} \mathbf{u}_b \right] + \frac{\partial g_b}{\partial \psi_p}, \end{aligned}$$

with $t_2 = -\frac{1}{2}[\text{tr}(\mathbf{R}^{-1}\ddot{\mathbf{R}}) - \text{tr}(\mathbf{A}^2)]$. The (ψ_p, ψ_p) entry follows from differentiating (14) once more, using $\partial_\phi \text{tr}(\mathbf{R}^{-1}\dot{\mathbf{R}}) = \text{tr}(\mathbf{R}^{-1}\ddot{\mathbf{R}}) - \text{tr}(\mathbf{A}^2)$ and $\partial_\phi(\mathbf{R}^{-1}\dot{\mathbf{R}}\mathbf{R}^{-1}) = \mathbf{R}^{-1}(\ddot{\mathbf{R}} - 2\dot{\mathbf{R}}\mathbf{R}^{-1}\dot{\mathbf{R}})\mathbf{R}^{-1}$. These match exactly the temporal-range derivatives already used in the spatio-temporal code of SDALGCP, with ϕ in place of the Matérn range. The asymptotic covariance of $\hat{\boldsymbol{\theta}}$ is $(-\nabla^2 \ell_{\text{MC}})^{-1}$ at the maximiser, so the direct method yields a genuine standard error for ϕ .

5.3 Numerical verification

At an off-optimum test point the analytic gradient and Hessian of (11) agree with `numDeriv` to 8×10^{-9} and 5×10^{-8} respectively; the Hessian is symmetric to machine precision. These checks run as unit tests (`tests/testthat/test-direct.R`).

6 Grid versus direct: practical guidance

	grid (profile)	direct (continuous)
ϕ estimate	restricted to grid nodes	continuous (no discretisation error)
SE for ϕ	not available	from the joint Hessian
profile shape	fully visible (good for multimodality)	not traced
cost	one MCML fit per grid node	one joint optimisation
kernel	any κ	exponential ($\kappa = \frac{1}{2}$) for now

On a 9×9 simulated lattice the two agree on (β, σ^2) and on ϕ to within the direct method's standard error, while `direct` is faster than even a 15-node grid and removes the grid-boundary artefact in $\hat{\phi}$. We keep **both**: `grid` as the robust default (it exposes the whole profile and supports any κ), `direct` for speed and an honest ϕ standard error. They are selected by `phi_method = c("grid", "direct")` in `SDALGCP2()` / `mcml_fit()`.

References. Christensen, O.F. (2004). Monte Carlo maximum likelihood in model-based geo-statistics. *JCGS* 13(3), 702–718.

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